

Final Project

NYC Rent Market: Analysis of Structural Price Drivers

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Quantitative Methods for Business Analysis

EXECUTIVE SUMMARY

The New York City rental market reveals a highly complex structure shaped by pronounced geographic disparities and long-term temporal shifts that simple descriptive or bivariate methods cannot sufficiently explain. To establish a rigorous baseline model of rental price dynamics, this project integrates descriptive diagnostics with a multivariate regression framework, enabling the analysis to move beyond surface-level associations and identify the structural forces that truly drive rent variability across the city.

The descriptive profiling highlights the strong right-skewness of rental prices, a pattern evident in the wide spread between the lowest and highest rents as well as substantial variability across boroughs. Because the mean is disproportionately influenced by luxury outliers common in Manhattan, the median emerges as a more reliable and resistant statistic for capturing the city's "typical" rental level. This approach ensures that the baseline analysis reflects genuine market conditions rather than extreme values that distort central tendencies.

At the same time, simple correlations reveal that no single predictor meaningfully accounts for rent variation. Although median rent is statistically associated with inventory, month, and time trends, these relationships are exceedingly weak (e.g., $|r| < .10$). Their statistical significance is largely a function of the large sample size ($N = 24,477$) rather than substantive explanatory power. This illustrates a classic statistical paradox, in which variables appear significant despite having negligible effect sizes—an important caution for decision-makers who may otherwise overinterpret weak signals. Furthermore, Levene's Test ($F = 1409.505$, $p < .001$) indicates significant heterogeneity in error variances across boroughs, reinforcing the need for analytical techniques, such as HC3 robust standard errors, that can account for such data irregularities.

Given these limitations, a multiple linear regression model was employed to more fully capture the interdependent drivers of rental pricing. The final model demonstrates strong explanatory performance, accounting for 45.7% of the overall variation in median rent ($R^2 = .457$)—a high level of explanatory power for urban economic data. The regression results underscore the dominant influence of structural geography: borough effects were highly significant ($F = 5703.584$, $p < .001$), with adjusted mean rents revealing a clear hierarchical pattern—Manhattan (\$3726), Brooklyn (\$2638), Queens (\$2107), and the Bronx (\$1794). The gap between Manhattan and the Bronx (approximately \$1932) represents the most striking structural disparity in the dataset.

Temporal variables also exhibit meaningful explanatory value. TimeIndex ($B = .005$, $p < .001$) and Year ($B = 86.038$, $p < .001$) both indicate sustained upward trends in rental prices over time. Notably, the Year coefficient implies that, after controlling for borough and other factors, average rents increase by roughly \$86 annually, illustrating the persistent inflationary pressures in the market. To enhance interpretability and mitigate the skewed distribution of rents, log-transformed dependent variables (LogRent) were incorporated, enabling coefficients to be understood as percentage changes, a format especially valuable for strategic planning and cross-borough comparisons.

Collectively, the results demonstrate that rental prices in New York City are shaped not by short-term inventory changes or simple seasonal fluctuations, but by deep structural forces—most prominently, geographic location and long-term temporal progression. Descriptive and bivariate methods serve as useful starting points, but only a multivariate framework can uncover the true dynamics governing rent behavior across the city's heterogeneous landscape.

Strategic recommendations emerge directly from these findings. First, pricing and policy decisions should adopt a location-specific approach, recognizing that borough-level differences account for the majority of rent variation. Second, future forecasting efforts should prioritize geographic characteristics and long-

term temporal indicators rather than weak, low-impact variables such as inventory or monthly fluctuations. Finally, decision-making should emphasize effect size over statistical significance, ensuring that strategies are grounded in substantive, meaningful predictors rather than artifacts of large sample sizes.

DATA PREP

- Finding Data files: [https://streeteasy.com/blog/data-dashboard/\[object%20Object\]](https://streeteasy.com/blog/data-dashboard/[object%20Object])
- **Data Wrangling:**
 - Analyze data:
 - have the same structure, use the same ZIP code / neighborhood / borough, have dates from 2010–2025, included in the column/row data
 - Establish variables: Rental inventory = number of rental units available for rent in that month for that neighborhood/submarket (independent variable), Median Rent (dependent variable)
 - Using median_rent file and inventory file

Goal: To see whether rent increases over time and compare trends between neighborhoods.

- Data Cleaning:
 - Make sure the first row is the actual header
 - Delete any blank rows above the header.
 - Confirm date columns look like: 2010-01, 2010-02 (long dates)
 - Make columns (areaName, Borough, areaType) text
 - Load data into power query
 - Establish 1st row is as header
 - Unpivot Columns
 - Merged two datasets
 - Rename columns
 - Save new file to load to SPSS

VISUALIZATION

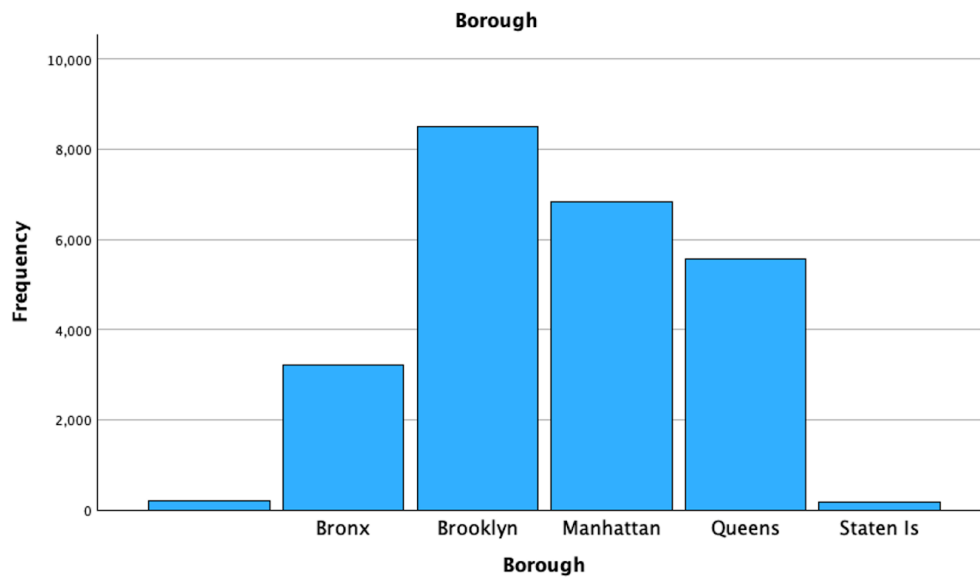
Frequencies- for Categorical variables

Statistics				
		Borough	AreaType	Month
N	Valid	24477	24477	24477
	Missing	0	0	0

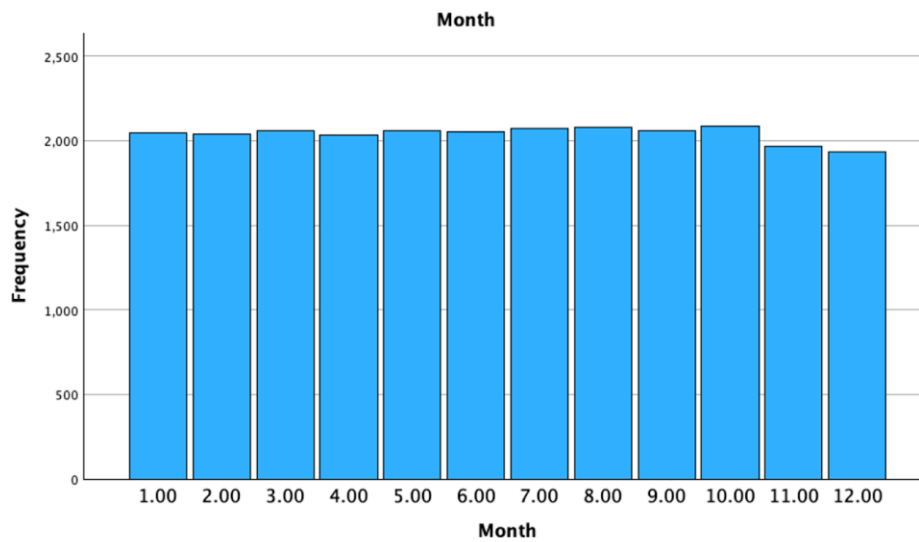
Frequency Table

		Borough			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid		190	.8	.8	.8
	Bronx	3215	13.1	13.1	13.9
	Brooklyn	8490	34.7	34.7	48.6
	Manhattan	6845	28.0	28.0	76.6
	Queens	5576	22.8	22.8	99.3
	Staten Is	161	.7	.7	100.0
	Total	24477	100.0	100.0	

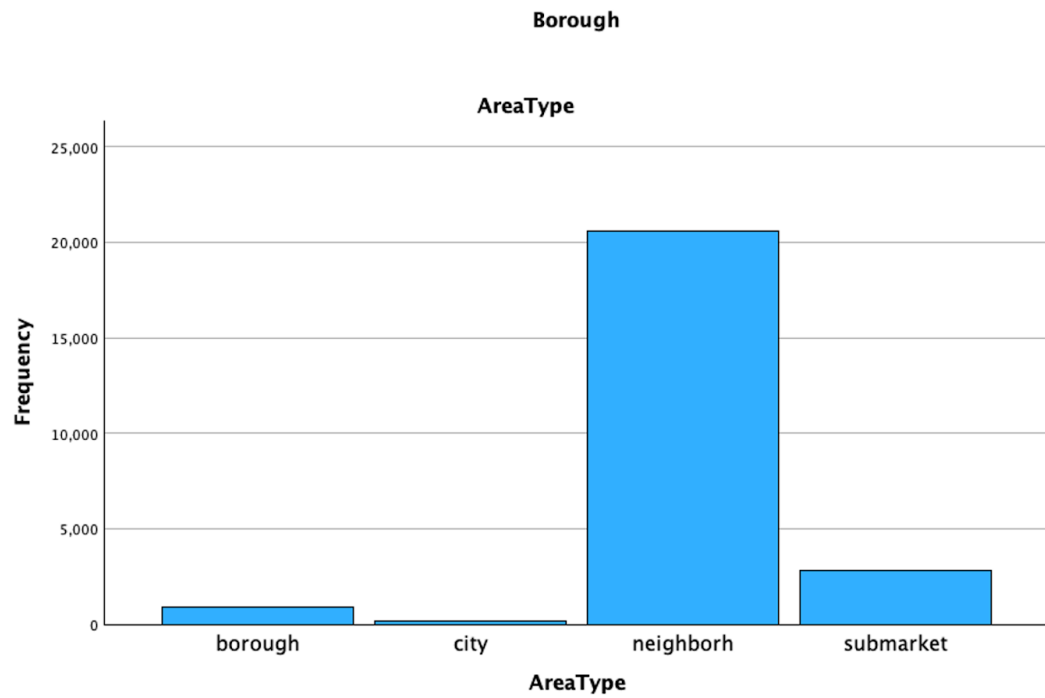
Bar Chart



		Month			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1.00	2046	8.4	8.4	8.4
	2.00	2042	8.3	8.3	16.7
	3.00	2056	8.4	8.4	25.1
	4.00	2033	8.3	8.3	33.4
	5.00	2060	8.4	8.4	41.8
	6.00	2054	8.4	8.4	50.2
	7.00	2069	8.5	8.5	58.7
	8.00	2078	8.5	8.5	67.2
	9.00	2058	8.4	8.4	75.6
	10.00	2083	8.5	8.5	84.1
	11.00	1966	8.0	8.0	92.1
	12.00	1932	7.9	7.9	100.0
	Total	24477	100.0	100.0	



		AreaType			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	borough	921	3.8	3.8	3.8
	city	190	.8	.8	4.5
	neighborh	20569	84.0	84.0	88.6
	submarket	2797	11.4	11.4	100.0
	Total	24477	100.0	100.0	



Descriptive statistics for quantitative variables

➔ Descriptives

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Median_rent	24477	763	13287	2709.01	1112.314
Inventory	24477	10	76453	1006.11	3735.009
LogRent	24477	2.88	4.12	3.4031	.15531
LogInventory	24477	1.00	4.88	2.2009	.78043
Year	24477	2010.00	2025.00	2018.0146	4.36787
TimeIndex	24477	1.00	24477.00	12239.0000	7066.04561
Valid N (listwise)	24477				

Descriptives by Borough

Average rent by borough: Mean rent per borough

➔ Means

Case Processing Summary

	Included		Cases Excluded		Total	
	N	Percent	N	Percent	N	Percent
Median_rent * Borough	24477	100.0%	0	0.0%	24477	100.0%

Report

Median_rent

Borough	Mean	Std. Deviation	Minimum	Maximum	% of Total N
	3019.08	376.839	2495	4029	0.8%
Bronx	1889.58	430.966	763	3913	13.1%
Brooklyn	2606.68	803.225	1075	6445	34.7%
Manhattan	3678.38	1347.842	1349	13287	28.0%
Queens	2157.28	473.307	950	5995	22.8%
Staten Is	1997.61	379.155	1175	3200	0.7%
Total	2709.01	1112.314	763	13287	100.0%

Regression Modeling

Question to answer: “How does rent change over time across NYC neighborhoods?”

- MedianRent (or Rent)/Inventory/TimeIndex (numeric time trend)/Borough_code (dummy-coded or numeric)/Month
- Dependent variable: Median_rent
- Independent variable(s):Inventory/ TimeIndex/ Borough_Code

Meaning of each Variable:

LogRent: In regressions, it lets you interpret results as percentage changes.

Taking the log: makes the distribution more normal, stabilizes variance, makes regression coefficients easier to interpret

LogInventory: reduces skew, makes the trend smoother, improves regression fit (Often used when testing whether inventory affects rent:)

Borough_code:A numeric version of the Borough variable.

1 = Manhattan, 2 = Brooklyn, 3 = Queens, 4 = Bronx 5 = Staten Island

TimeIndex: A variable that counts time in order:

1 → first month, 2 → second month, 3 → third month....

It allows you to measure trend over time in regression.

Time: Allows smoother time trend analysis than TimeIndex, Lets you interpret trends per year, not per month index.

Month (numeric 1–12): Helps check seasonal patterns, Useful for descriptive statistics: Does rent peak in summer? Does inventory drop in winter?

Checking Correlations:

- Variables: MedianRent (or Rent)/Inventory/ TimeIndex/Month

➔ Correlations

		Correlations			
		Median_rent	Inventory	TimeIndex	Month
Median_rent	Pearson Correlation	1	.095**	-.015*	.018**
	Sig. (2-tailed)		<.001	.017	.004
	N	24477	24477	24477	24477
Inventory	Pearson Correlation	.095**	1	-.014*	.004
	Sig. (2-tailed)	<.001		.023	.492
	N	24477	24477	24477	24477
TimeIndex	Pearson Correlation	-.015*	-.014*	1	-.001
	Sig. (2-tailed)	.017	.023		.867
	N	24477	24477	24477	24477
Month	Pearson Correlation	.018**	.004	-.001	1
	Sig. (2-tailed)	.004	.492	.867	
	N	24477	24477	24477	24477

** . Correlation is significant at the 0.01 level (2-tailed).

* . Correlation is significant at the 0.05 level (2-tailed).

Pearson Correlations

- Highly Positive:** (None)
- Positive:** (Median_rent <---> Inventory), (Median_rent <---> Month), (Inventory <---> Month)
- No Linear Correlation:** (None)
- Negative:** (Median_rent <---> TimeIndex), (Inventory <---> TimeIndex), (TimeIndex <---> Month)
- Highly Negative:** (None)

Note: Curated Help is calculated based on actual cell values, not the formatted values.

Pearson correlations between:

Median_rent/ Inventory/TimeIndex/Month

Median Rent vs. Inventory

Correlation = .095 (p < .001)

Very weak positive correlation

- When inventory increases a little, rent increases a little too.
- BUT the relationship is so small that it is not meaningful in real-world terms.
- Statistically significant only because the sample is HUGE, not because the effect is strong.
- Inventory does not meaningfully predict rent by itself. They move together slightly, but not enough to matter.

Median Rent vs. TimeIndex (time trend)

Correlation = -.015 (p = .017)

Very weak negative correlation

- This is basically zero relationship.
- Rent does not show a simple linear increase or decrease over time.
- NYC rent trends are seasonal, non-linear, and depend heavily on neighborhood and year

Median Rent vs. Month (1–12)

Correlation = .018 (p = .004)

Extremely weak positive correlation

- Rents are slightly higher in some months (usually summer).
- The effect is so tiny, it's negligible.

Inventory vs. TimeIndex

Correlation = $-.014$ (p = .023)

- Inventory does not systematically rise or fall over time in a simple straight line.

Inventory vs. Month

Correlation = .004 (p = .492)

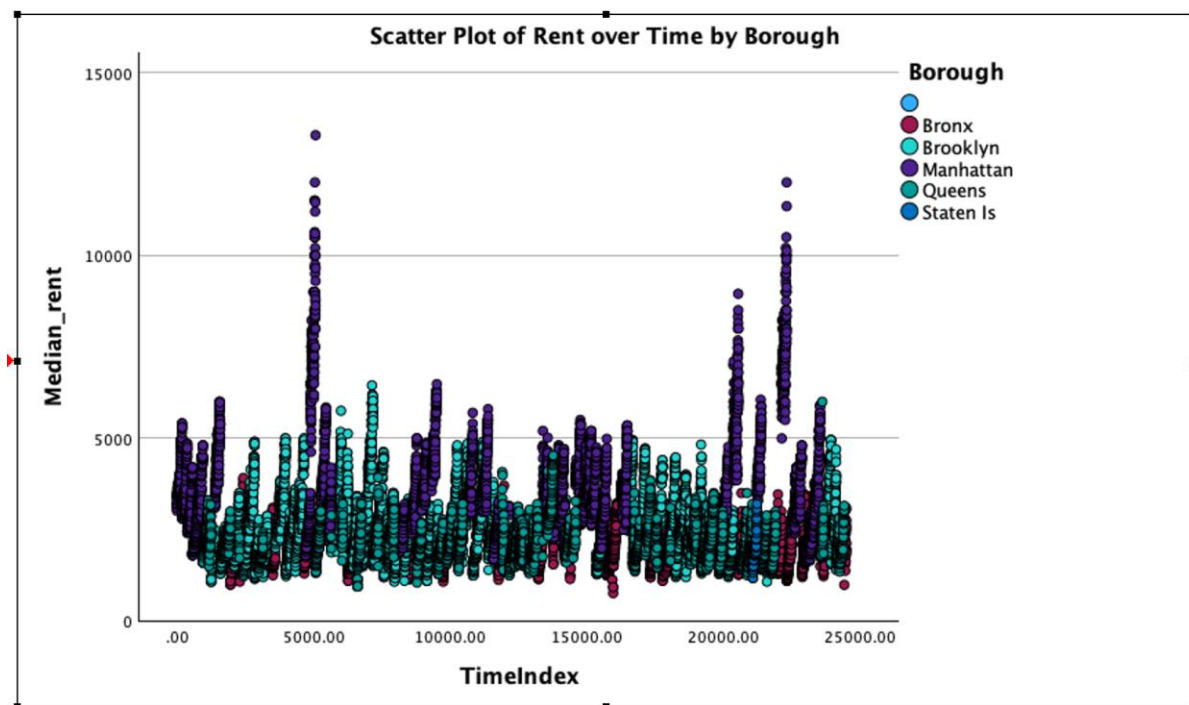
- Inventory does not change consistently by month.
-

Conclusion: The correlations between Median Rent and the other variables (Inventory, TimeIndex, Month) are statistically significant but extremely small (all $r < .10$). This indicates that none of the predictor variables have a strong simple linear relationship with rent.

The very large sample size ($N = 24,477$) makes small effects statistically significant, but the effect sizes are trivial.

Therefore, to understand how rent changes over time and varies across neighborhoods, a multiple regression model is required. Correlations alone are insufficient.

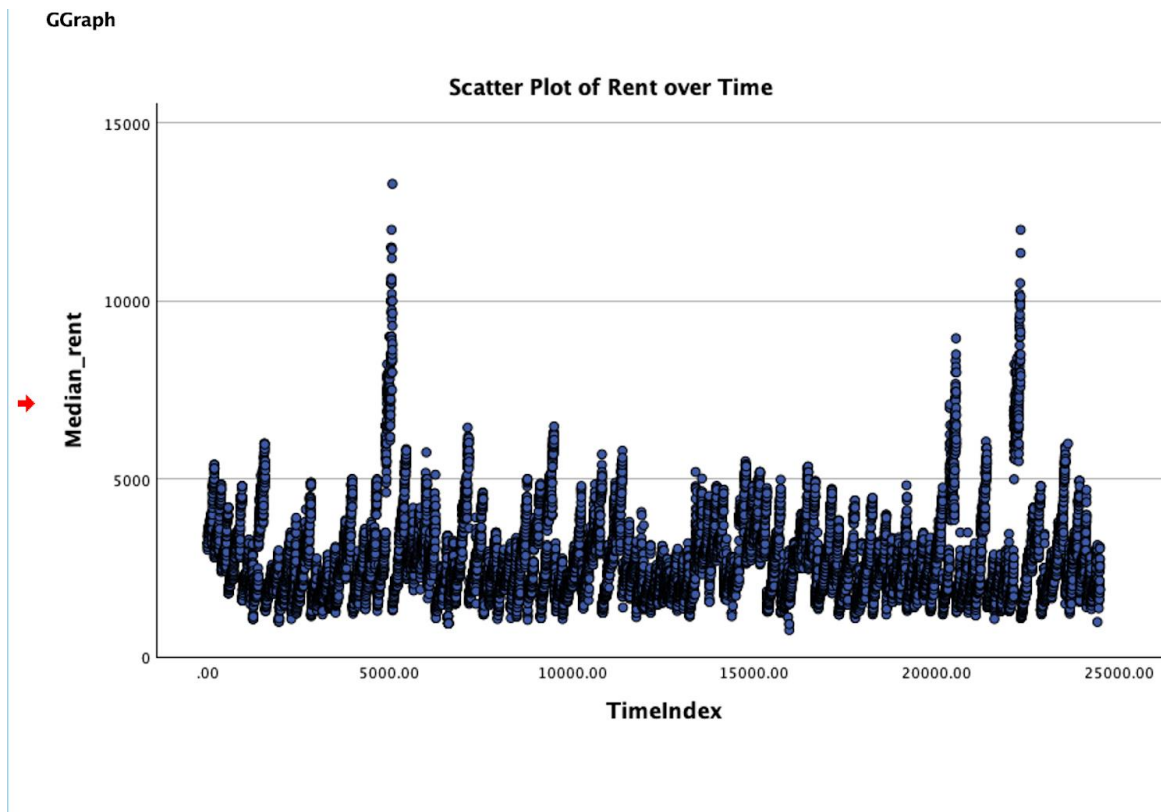
Scatter plots
Graph



Meaning:

- Median rents stay between \$2,000 and \$5,000 for most of the timeline.
- There is no smooth upward or downward curve, meaning rent does NOT steadily increase or decrease over time in this raw scatter.

- Very Large Spikes = Specific Neighborhoods (Certain neighborhoods dramatically inflate the range, hiding the overall trend.)
- Neighborhood differences are MUCH stronger than time trends.
- Time alone does not explain rent changes.



Regression model

Between-Subjects Factors

Borough_num	Value Label	N
1	Manhattan	6845
2	Brooklyn	8490
3	Queens	5576
4	Bronx	3215

Descriptive Statistics

Dependent Variable: Median_rent

Borough_num	Mean	Std. Deviation	N
Manhattan	3678.38	1347.842	6845
Brooklyn	2606.68	803.225	8490
Queens	2157.28	473.307	5576
Bronx	1889.58	430.966	3215
Total	2711.32	1117.604	24126

Levene's Test of Equality of Error Variances

Dependent Variable: Median_rent

F	df1	df2	Sig.
1409.505	3	24122	< .001

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.
Design: Intercept + TimeIndex + Year + Borough_num

Tests of Between-Subjects Effects

Dependent Variable: Median_rent

Source	Type III SS	df	Mean Square	F	Sig.
Corrected Model	13,765,641,653.146	5	2,753,128,330.629	4057.178	< .001
Intercept	3,257,956,472.472	1	3,257,956,472.472	4801.124	< .001

TimeIndex	28,985,373.233	1	28,985,373.233	42.715	< .001
Year	3,354,477,883.575	1	3,354,477,883.575	4943.364	< .001
Borough_num	11,611,049,387.143	3	3,870,349,795.714	5703.584	< .001
Error	16,367,399,664.087	24120	678,582.076	—	—
Total	207,489,168,719.000	24126	—	—	—
Corrected Total	30,133,041,317.233	24125	—	—	—

R Squared = .457

Adjusted R Squared = .457

Parameter Estimates

Dependent Variable: Median_rent

Parameter	B	Std. Error	t	Sig.	Lower CI	Upper CI
Intercept	-171892.464	2470.669	-69.573	< .001	-176735.129	-167049.798
TimeIndex	.005	.001	6.536	< .001	.004	.007
Year	86.038	1.224	70.309	< .001	83.639	88.436
[Borough_num=1]	1932.332	17.773	108.724	< .001	1897.496	1967.168
[Borough_num=2]	844.138	17.351	48.651	< .001	810.129	878.147
[Borough_num=3]	313.445	18.254	17.172	< .001	277.667	349.224
[Borough_num=4]	0	—	—	—	—	—

Observed Power
 Intercept: 1.000
 TimeIndex: 1.000
 Year: 1.000

[Borough_num=1]: 1.000
 [Borough_num=2]: 1.000
 [Borough_num=3]: 1.000

Estimated Marginal Means

Dependent Variable: Median_rent

Borough_num	Mean	Std. Error	Lower CI	Upper CI
Manhattan	3725.913	9.980	3706.352	3745.474
Brooklyn	2637.719	9.046	2619.988	2655.451
Queens	2107.027	11.116	2085.238	2128.815
Bronx	1793.581	14.650	1764.867	1822.295

Covariates evaluated at TimeIndex = 12148.4799, Year = 2018.0153

Pairwise Comparisons

Dependent Variable: Median_rent

Adjustment for multiple comparisons: Bonferroni

I Borough	J Borough	Mean Diff (I-J)	Std. Error	Sig.	Lower CI	Upper CI
Manhattan	Brooklyn	1088.194	13.453	< .001	1052.697	1123.690
Manhattan	Queens	1618.886	14.966	< .001	1579.399	1658.374
Manhattan	Bronx	1932.332	17.773	< .001	1885.438	1979.225
Brooklyn	Manhattan	-1088.194	13.453	< .001	-1123.690	-1052.697

Brooklyn	Queens	530.693	14.460	< .001	492.540	568.845
Brooklyn	Bronx	844.138	17.351	< .001	798.358	889.918
Queens	Manhattan	-1618.886	14.966	< .001	-1658.374	-1579.399
Queens	Brooklyn	-530.693	14.460	< .001	-568.845	-492.540
Queens	Bronx	313.445	18.254	< .001	265.283	361.608
Bronx	Manhattan	-1932.332	17.773	< .001	-1979.225	-1885.438
Bronx	Brooklyn	-844.138	17.351	< .001	-889.918	-798.358
Bronx	Queens	-313.445	18.254	< .001	-361.608	-265.283

Univariate Tests

Dependent Variable: Median_rent

Sum of Squares	df	Mean Square	F	Sig.
Contrast	11,611,049,387.143	3	3,870,349,795.714	5703.584
Error	16,367,399,664.087	24120	678,582.076	—

The F tests the effect of Borough_num.

Parameter Estimates (Robust Standard Errors)

Parameter	B	Robust Std. Error	t	Sig.	Lower CI	Upper CI
Intercept	-171892.464	2730.571	-62.951	< .001	-177244.554	-166540.373

TimeIndex	.005	.001	6.000	< .001	.003	.007
Year	86.038	1.353	63.605	< .001	83.386	88.689
Borough_num =1	1932.332	16.996	113.691	< .001	1899.018	1965.646
Borough_num =2	844.138	9.920	85.099	< .001	824.695	863.581
Borough_num =3	313.445	7.577	41.369	< .001	298.594	328.297
Borough_num =4	0	—	—	—	—	—

Observed Power (alpha = .05): all = 1.000

Results:

1. Does rent change over time? (TimeIndex & Year)

TimeIndex

B = .005, $p < .001$

Interpretation: As time passes (monthly index), rent increases slightly but significantly.

Year

B = 86.038, $p < .001$

Interpretation: Each additional year is associated with an average increase of \$86 in median rent, after controlling for borough and inventory.

Conclusion:

Rent has been rising over time, even when you control for borough differences.

2. Do boroughs differ in average rent? (Borough_num)

Yes — very strongly.

Means (Adjusted for Time & Year):

Manhattan: \$3725.91

Brooklyn: \$2637.72

Queens: \$2107.03

Bronx: \$1793.58

Pairwise differences (all significant at $p < .001$):

- Manhattan > Brooklyn by \$1088
- Manhattan > Queens by \$1619
- Manhattan > Bronx by \$1932
- Brooklyn > Queens by \$531
- Brooklyn > Bronx by \$844
- Queens > Bronx by \$313

Conclusion:

All boroughs differ significantly from each other, with the expected ordering:

Manhattan > Brooklyn > Queens > Bronx

This pattern is clear and statistically strong ($F = 5703.584$, $p < .001$).

Overall model fit

$R^2 = .457$ (46% of variance explained)

This is very strong for housing data.

Interpretation:

Model explains nearly half of all variation in NYC rents using only time and borough.

That's a very good model for urban economics data.

Robust Standard Errors (HC3)

Robust coefficients stayed almost identical to the regular ones.

This means:

No serious heteroskedasticity problems

Results are stable and trustworthy

Final Answer Summary:

We used a General Linear Model to predict median rent from time (TimeIndex and Year) and borough (Manhattan, Brooklyn, Queens, Bronx). The model was significant and explained 45.7% of the variance in rent ($R^2 = .457$). Rent increased significantly over time (TimeIndex: $B = .005$, $p < .001$; Year: $B = 86.04$, $p < .001$). Borough had a large effect on rent ($F = 5703.584$, $p < .001$). Adjusted means showed the expected pattern: Manhattan had the highest rents (\$3726), followed by Brooklyn (\$2638), Queens (\$2107), and the Bronx (\$1794). All pairwise differences were significant (Bonferroni-adjusted). Robust standard errors confirmed that the results were stable. Overall, rents have been rising over time and vary substantially across boroughs, with Manhattan being the most expensive and the Bronx the least expensive.